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August 25, 2026

To Whom It May Concern

This is to certify that Mr. Sahil Thakur (24JE0573), a third year B.Tech. student of the Department of Chemical Engineering at Indian Institute of Technology (Indian School of Mines) Dhanbad has successfully completed his summer research internship under my guidance in the Department of Chemical Engineering, IIT Ropar. The internship period was carried out from 15 May 2026 to 8 July 2026.

Mr. Sahil carried out the design and synthesis of Zn-, Cu-, Co- and Cr-modified HZSM-5 catalysts, with particular emphasis on Zn-based bimetallic systems as promising catalysts for converting used cooking oil (UCO) into valuable aromatic hydrocarbons. He is now familiar with catalyst synthesis methods (wet impregnation), pelletization and preparation for fixed bed reactor studies. During his internship, he actively participated in research work and completed all assigned objectives satisfactorily.

Sincerely,



Dr. Navin Gopinathan
Assistant Professor
Department of Chemical Engineering
IIT Ropar